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MA3521 — Missing Eigenvectors

Name:

$$\mathbf{y}' = \boldsymbol{A}\mathbf{y}, \quad \boldsymbol{A} = \begin{pmatrix} -2 & 2 \\ -8 & 6 \end{pmatrix}, \quad \lambda_1 = \lambda_2 = 2, \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \mathbf{y}(0) = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$$

(a) Write down the form of the complementary solution. Use $\mathbf{w}$ for the additional vector.

$\mathbf{y}_c(t) =$

(b) You row reduce $\boldsymbol{M}_w$ to find $\mathbf{w}$.

$\boldsymbol{M}_w =$

(c) You row reduce $\boldsymbol{M}_{\text{IC}}$ to satisfy the initial condition.

$\boldsymbol{M}_{\text{IC}} =$

$$\mathbf{y}' = \boldsymbol{A}\mathbf{y}, \quad \boldsymbol{A} = \begin{pmatrix} -3 & 1 & 5 \\ -4 & 1 & 10 \\ 0 & 0 & 4 \end{pmatrix}, \quad \lambda_{1,2} = -1,\ \lambda_3 = 4, \quad \mathbf{v}_1 = \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix},\ \mathbf{v}_3 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad \mathbf{y}(0) = \begin{pmatrix} 3 \\ 7 \\ 2 \end{pmatrix}$$

(a) Write down the form of the complementary solution. Use $\mathbf{w}$ for the additional vector.

$\mathbf{y}_c(t) =$

(b) You row reduce $\boldsymbol{M}_w$ to find $\mathbf{w}$.

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